

# CINEVERSE — ROLE & FUNCTIONALITY MATRIX

Complete Feature Breakdown Across Customer, Theatre Manager, Theatre Staff, Super Admin & Core Infrastructure

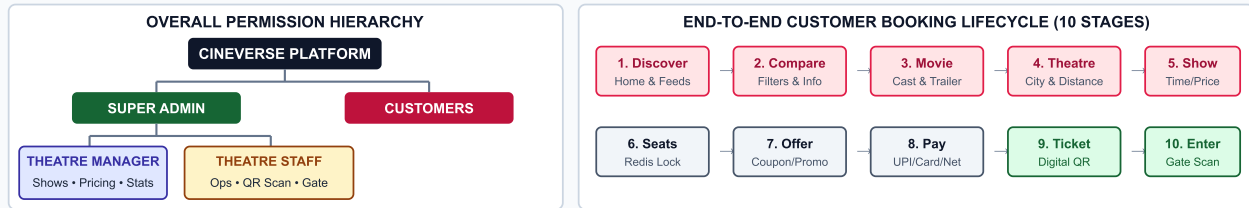


Figure 1.1: CineVerse Four-Role Governance Hierarchy & Ten-Stage Customer Booking Funnel

## 1. Customer / End User — Modules A to F (Onboarding & Discovery)

ROLE 1 • LARGEST  
MODULE

- **A. Registration, Authentication & Security:**
  - **Account Creation:** Register using Name, Email, Mobile Number, Password, Confirm Password, and mandatory Terms & Conditions acceptance.
  - **Multi-Channel Login:** Authenticate via Email + Password, Mobile + OTP, or Google OAuth; includes Forgot Password, Reset Password, and Logout flows.
  - **Platform Security:** Enforces password hashing, JWT/session management, refresh tokens, OTP expiry timers, failed login attempt protection, automatic account lock after repeated failures, and multi-device session management.
- **B. User Profile & Personalization Preferences:**
  - **Personal Information:** View profile and edit Name, Profile Photo, Email, Mobile Number, and Password.
  - **Stored Preferences (Recommendation Fuel):** Persist Preferred Language, Preferred Cinema, Preferred Genres, Preferred Seat Type, Preferred Show Timing, and Notification Channel Preferences to power personalized discovery.
- **C. Location & Proximity Engine:**
  - **Geolocation Controls:** Select city manually, auto-detect approximate user location, switch active location, save favourite city, and view nearby theatres (e.g., **Belagavi** → **Movies near you**).
  - **Distance Sorting:** Computes `distance(user, theatre)` dynamically to rank and sort cinemas by proximity.
- **D. Home Page, Curated Feeds, Filters & Global Search:**
  - **Movie Discovery Sections:** Browse *Now Showing*, *Coming Soon*, *Popular*, *Trending*, *Top Rated*, *Recommended For You*, and *New Releases*.
  - **Multi-Faceted Filters:** Narrow listings by Language, Genre, Rating, Date, Format (2D/3D/IMAX), Price Range, and Distance.
  - **Unified Search Bar:** Instant search across Movie Title, Actor, Director, Theatre Name, Genre, and Language.
- **E. Movie Details & Viewer Actions:**
  - **Comprehensive Metadata:** Displays Official Poster, Title, User/Critic Rating, Duration, Censor Certificate, Language, Genre, Release Date, Director, Cast List, Synopsis, and embedded Trailer.
  - **Interactive Actions:** Watch trailer, Add to Watchlist, Rate movie, Write/Read movie reviews, Share movie link, and proceed to *Book Tickets*.
- **F. Personalized Movie Recommendation Engine:**
  - **Algorithmic Signals:** Generates tailored suggestions based on Previous Bookings, Watched Movies, Genres, Favourite Actors, Directors, User Ratings, Language, Location, and Local Popularity.
  - **Contextual Output:** Surfaces explainable prompts such as *"Because you watched Interstellar..."* → recommends *Dune*, *Oppenheimer*, and *Arrival*.

# 1. Customer / End User — Modules G to O (Theatres, Seats & Checkout)

ROLE 1 • CORE  
BOOKING

- G. Theatre Discovery & Showtime Matrix:**
  - Selection Hierarchy:** Follows **Movie** → **City** → **Date** → **Theatres**; each listing displays Theatre Name (e.g., *PVR Belagavi*), Star Rating (★ 4.4), Distance (2.1 km), Screen Count (4 Screens), and clickable showtime pills (10:30 AM, 1:30 PM, 4:30 PM, 7:30 PM).
  - Theatre Filters:** Filter available cinemas by Distance, Rating, Price Tier, Amenities, Format, Language, and Show Time.
- H. Theatre Details & Facility Profile:**
  - Venue Information:** Presents Theatre Name, Full Address, Interactive Map, Contact Info, User Rating, Interior Photos, and Screen Count.
  - Verified Amenities:** Highlights Parking, Wheelchair Access, Dolby Atmos / Projection Formats, Food & Beverages availability, and Recliner Seats.
- I. Show Selection & Real-Time Pricing Preview:**
  - Drill-Down Flow:** **Date** → **Theatre** → **Screen** → **Show**; surfaces Start Time, End Time, Audio Language, Visual Format, Available Seats, and Live Occupancy Percentage (e.g., 7:30 PM • Normal ₹180 | Premium ₹250 | Recliner ₹350 • Available: 73%).
- J. Interactive Seat Selection Grid:**
  - Visual Auditorium Map:** Renders exact screen orientation and row/column layout (Rows A–E+) with five distinct seat statuses: *Available*, *Booked*, *Selected*, *Temporarily Locked*, and *Unavailable / Broken*.
- K. Smart Seat Recommendation Algorithm:**
  - Automated Best-Seat Finder:** Users can click "Find Best Seats" and choose a preference profile—*Best View*, *Couple*, *Cheapest*, *Near Exit*, *Center*, or *Accessible*—to automatically highlight optimal contiguous seats.
- L. Real-Time Seat Locking & Concurrency Hold:**
  - Countdown Hold:** Selecting seats (e.g., C7, C8) triggers an immediate temporary lock with a live countdown (Time remaining: 04:32); all other browsing users instantaneously see C7 🔒, C8 🔒. If payment is not completed before timer expiry, seats auto-release back to the pool.

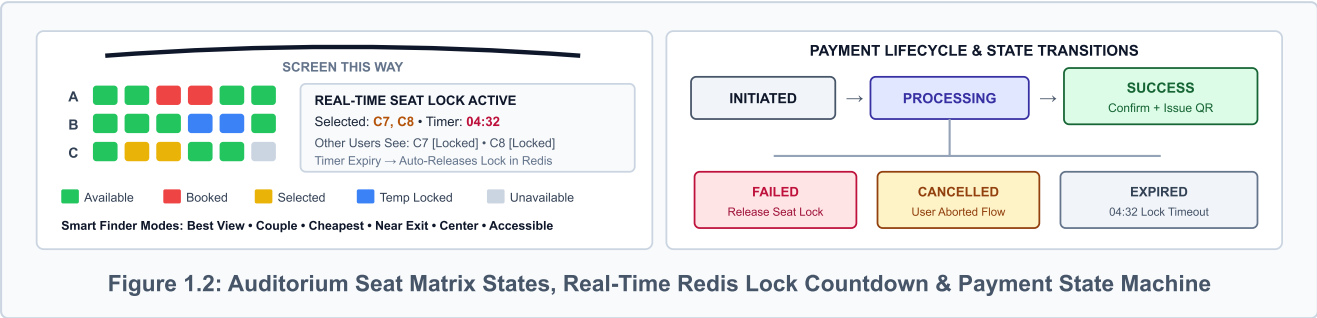


Figure 1.2: Auditorium Seat Matrix States, Real-Time Redis Lock Countdown & Payment State Machine

- M. Booking Cart & Checkout Breakdown:**
  - Itemized Summary:** Displays Movie, Theatre, Screen, Date, Time, Selected Seats (C7, C8), Base Ticket Price, Convenience Fee, Taxes, Applied Discount, and Final Payable Total. Supports applying Coupons, Offers, Wallet Balance, and selecting Payment Method.
- N. Coupon & Promo Validation Engine:**
  - Customer Controls & Rules:** Enter custom coupon code, browse available coupons, apply or remove coupon. Backend validates Expiry Date, Minimum Booking Amount, Maximum Discount Cap, Eligible Movie, Eligible Theatre, Eligible User Tier, and Per-User/Global Usage Limits.
- O. Multi-Method Payment Processing:**
  - Supported Rails & States:** Supports UPI, Credit/Debit Card, Net Banking, and Digital Wallets across deterministic states: **INITIATED** → **PROCESSING** → **SUCCESS** (or **FAILED** / **CANCELLED** / **EXPIRED**).

# 1. Customer / End User — Modules P to X (Post-Booking, AI & Security)

ROLE 1 • POST-PURCHASE

- P & Q. Booking Confirmation & Digital QR Ticket:**
  - Instant Confirmation:** Displays Booking ID (**CV123456**), Movie (*Interstellar*), Theatre (*XYZ Cinema*), Screen (Screen 2), Seats (**C7, C8**), Date (**10 Oct 2026**), and Time (**7:30 PM**).
  - Digital Ticket Actions:** Generates scannable QR Code with Payment Status; users can View, Download PDF/Pass, Share, or Add to Wallet/Calendar.
- R & S. My Bookings, Cancellation Policy & Refund Tracking:**
  - Booking Lifecycle Tabs:** Organized into *Upcoming*, *Completed*, and *Cancelled* tabs with options to View/Download Ticket, Cancel, Reschedule (if supported), View Payment Receipt, and Track Refund Status.
  - Transparent Refund Math:** Shows policy rules and exact deduction before confirmation (e.g., **Ticket Amount ₹500 – Cancellation Fee ₹50 = Refund ₹450**).
- T & U. Reviews, Ratings & Watchlist / Wishlist:**
  - Verified Post-Watch Feedback:** Rate movie, rate theatre, write text review, upload experience photos, and edit or delete own reviews.
  - Watchlist Alerts:** Add or remove movies from Watchlist; trigger automated alerts when the movie releases or when ticket bookings open.
- V & W. Omnichannel Notifications & AI Movie Booking Assistant:**
  - Event-Driven Alerts:** Dispatches across *In-App*, *Email*, *SMS*, and *Push* for Booking Confirmation, Payment Success/Failure, Show Reminder, Cancellation, Refund, Movie Release, Offers, and Seat Lock Expiry.
  - Conversational AI Assistant (Advanced Module):** Parses natural-language requests (e.g., *"I want 3 tickets for an action movie tomorrow evening under ₹700"*) and returns ready-to-book options (**Movie: WAR • Theatre: XYZ • 7:30 PM • Seats: C7, C8, C9 • Total: ₹660 → Book these seats?**).
- X. User Account Security & Privacy Controls:**
  - Session & Data Governance:** View active device sessions, execute *Logout from all devices*, change password, inspect login history, receive suspicious login notifications, configure privacy settings, and request permanent account deletion.

| Post-Booking Module      | Primary User Capabilities                     | System Validation / Logic                                | Reference Output Example                                     |
|--------------------------|-----------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------|
| Digital QR Ticket (P, Q) | View, Download, Share, Add to Calendar/Wallet | Cryptographic QR payload tied to Booking ID & seat state | ID: <b>CV123456</b> • Screen 2 • Seats C7, C8 • 7:30 PM      |
| Cancellations (R, S)     | Cancel ticket, inspect policy, track refund   | Cutoff window check & automatic fee deduction            | Paid <b>₹500</b> – Fee <b>₹50</b> = Net Refund <b>₹450</b>   |
| AI Assistant (W)         | Natural-language budget, genre & time query   | Filters showtimes + contiguous seats under price cap     | <b>WAR</b> • XYZ Cinema • C7–C9 • <b>Total ₹660</b> (≤ ₹700) |
| Notifications (V)        | Manage In-App, Email, SMS & Push toggles      | Async event bus triggered on state transitions           | Alerts on Lock Expiry, Reminder, Refund & Release            |

# 2. Theatre Representative / Manager — Modules A to G (Setup & Shows)

ROLE 2 • ASSIGNED THEATRE SCOPE

- A & B. Theatre Registration & Profile Management:**
  - Onboarding Workflow:** Submit Theatre Name, Address, Contact, GST/Business Information, Number of Screens, Amenities, Venue Images, and Legal Documents through status pipeline: **PENDING → ADMIN REVIEW → APPROVED / REJECTED**.
  - Profile Updates:** Edit Theatre Name, Address, Phone, Email, Description, Photos, Amenities, and Operating Hours (sensitive legal/location fields require Super Admin approval).
- C. Screen Management:**
  - Auditorium Setup:** Add new screen, edit specs, disable screen for maintenance, configure seating capacity, and assign projection/audio format (e.g., **Screen 1: Capacity 180, Format 2D**; **Screen 2: Capacity 220, Format IMAX**).

2. Theatre Representative / Manager — Modules H to Q (Pricing, Ops & Reports)

ROLE 2 • ANALYTICS & OPS

- H. Base & Dynamic Pricing Management:**
  - Tier & Rule Configuration:** Configure seat tier rates (e.g., **Regular ₹180 • Premium ₹250 • Recliner ₹350**) and apply dynamic rules for *Weekend, Holiday, Morning Show, Prime Time*, and *Demand-Based Pricing* within platform-enforced price ceilings.
- I, J & K. Live Occupancy Dashboard, Show-Level & Revenue Analytics:**
  - Real-Time Occupancy Monitor:** Tracks today's macro performance (**Tickets Sold: 482 • Revenue: ₹94,520 • Overall Occupancy: 76%**) alongside per-screen fill rates (**Screen 1: 82% • Screen 2: 71% • Screen 3: 74%**).
  - Show-Level Metrics:** Displays total Capacity, Sold Seats, Available Seats, Occupancy %, Gross Revenue, Cancellations, and Average Ticket Price (ATP) per show.
  - Revenue Visualizations:** Generates Daily, Weekly, and Monthly revenue trend graphs segmented by *Movie, Screen*, and *Showtime*.
- L. Theatre Booking Administration:**
  - Operational Oversight:** View, search, and filter theatre bookings; inspect customer details, assigned seats, and payment status; execute policy-compliant cancellations (managers are strictly prevented from altering completed payment ledger records).
- M. QR Scanning & Gate Entry Operations:**
  - Validation Pipeline:** Scan customer QR ticket → Booking Lookup → Validate Theatre & Screen → Check Show & Date → Verify Payment Status → Check Used Status → Permit Entry.
- N & O. Food & Beverage (F&B) Pre-Orders & Local Theatre Offers:**
  - Concession Catalogue (Optional):** Configure F&B items and prices (**Popcorn ₹150 • Coke ₹80 • Combo ₹250**) for customer pre-order during ticket checkout.
  - Local Promotions:** Launch theatre-specific deals (e.g., *Student Offer: 10% OFF, Tuesday Offer: ₹100 Tickets*), subject to Super Admin approval where required.
- P & Q. Theatre Staff Management & Exportable Reports:**
  - Staff Governance:** Add or remove theatre staff, assign operational permissions, disable staff accounts, and audit staff scan/login activity.
  - Standardized Reports:** Generate and export Daily Reports, Show Reports, Revenue Reports, Occupancy Reports, Cancellation Reports, and Ticket Sales Reports.

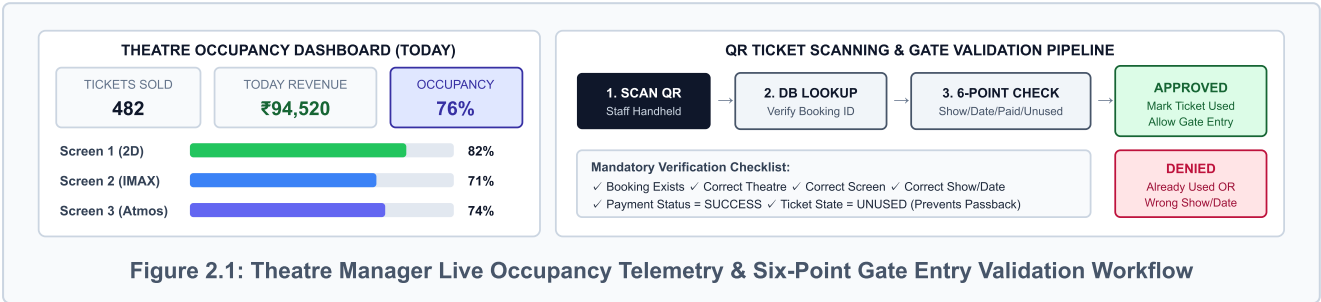


Figure 2.1: Theatre Manager Live Occupancy Telemetry & Six-Point Gate Entry Validation Workflow

3. Theatre Staff / Entry Operator — Restricted Operations Role

ROLE 3 • GATE & FLOOR OPS

- Intentional Role Restriction & Staff Authentication:**
  - Zero Financial/Config Access:** Theatre Staff accounts are strictly isolated from revenue analytics, ticket pricing, screen layouts, or catalogue management. Staff authenticate via assigned *Staff ID + Password* (with optional OTP verification).
- Today's Assigned Show Roster:**
  - Live Shift View:** Displays only today's active shows at the operator's theatre (e.g., **Screen 1 • 7:30 PM • Avengers**) for rapid gate preparation.
- QR Ticket Scanner & Instant Gate Decision:**
  - Automated Six-Point Verification:** Scans customer QR code to verify: (1) Booking exists, (2) Correct theatre, (3)

## 4. Super Admin (Platform Owner) — Modules A to L (Ecosystem Governance)

ROLE 4 • GLOBAL OWNER

- **A. Executive Admin Dashboard:**
  - **Macro Ecosystem Counters:** Real-time visibility into **Total Users (25,420)**, **Active Theatres (42)**, **Global Movies (180)**, **Today Bookings (3,284)**, **Today Revenue (₹8,42,500)**, and **Active Shows (310)**.
- **B & C. Global User Management & Theatre Lifecycle Governance:**
  - **User Controls:** View/search users, block or unblock accounts, suspend or delete accounts, reset credentials, inspect full booking history, and review suspicious activity flags.
  - **Theatre Controls & Approval (F):** Review submitted theatre applications, verify legal/GST documents, approve or reject onboarding (**Application** → **Admin Review** → **Docs Verified** → **Approved** → **Active**), suspend or reactivate theatres, and assign Theatre Managers.
- **D & E. Master Movie Catalogue & Global Data Integrity:**
  - **Exclusive Catalogue Authority:** Add, edit, or delete global movies; upload Official Poster and Trailer; manage Cast, Director, Genre, Language, Censor Certification, and Release Date. Theatre Managers are blocked from modifying official movie metadata to prevent platform-wide inconsistency.
- **G & H. Platform Pricing Rules & Global Coupon Engine:**
  - **Commercial Governance:** Set platform-wide Convenience Fees, Taxes, Maximum Discount Caps, Platform Commission Fees, Cancellation Policies, and Refund Rules.
  - **Global Promotions:** Create platform coupons (e.g., **CINE50 • 50% OFF • Min Spend ₹300 • Max Discount ₹100**) with granular rules for Expiry, Total Usage Limit, Per-User Limit, Theatre Eligibility, and Movie Eligibility.
- **I & J. Global Analytics & Real-Time Infrastructure Monitoring:**
  - **Ecosystem Intelligence:** Track platform-wide Revenue, Bookings, User Growth, Theatre Performance, Movie Performance, Occupancy %, Cancellations, and Refunds.
  - **System Health Telemetry:** Live status indicators (🟢 **Healthy** / 🔴 **Degraded**) for *API Server, Primary Database, Redis Cluster, Payment Service, Notification Service, and WebSocket Gateway*.
- **K & L. Immutable Audit Logs & Role-Based Access Control (RBAC):**
  - **Forensic Audit Trail:** Records every sensitive action with *Who, What, When, IP/Device, Previous Value, and New Value* (e.g., **ADMIN** → **Suspended Theatre #42** • **06 Oct 2026 21:32** • **Reason: Policy violation**).
  - **Granular Permission Mapping:** Replaces hardcoded role checks with RBAC permissions (**CREATE\_MOVIE, CREATE\_SHOW, MANAGE\_SEATS, SCAN\_TICKET, VIEW\_ANALYTICS, MANAGE\_USERS**) mapped across **SUPER\_ADMIN, THEATRE\_MANAGER, THEATRE\_STAFF, and CUSTOMER**.



Figure 3.1: Real-Time Seat Locking Architecture & Nine-Stage Transactional Booking State Machine

## 5. Cross-System Architecture & Failure Handling (Modules 1 to 8)

PLATFORM BACKBONE

- **1, 2 & 3. Unified RBAC Auth, Real-Time Seat Engine & Atomic Booking Transaction:**
  - **Cross-Role Security & Concurrency:** Strictly scoped JWT + RBAC middleware across all 4 roles; Redis-backed temporary seat locks paired with WebSocket broadcasts prevent double-booking during high-traffic blockbusters.
- **4. Deterministic Edge-Case & Failure Recovery Protocols:**
  - **Five Mandatory Resilience Guarantees:** (1) *Payment successful but server crashes:* Webhook/cron reconciliation matches gateway ledger and confirms booking; (2) *User closes browser:* Redis TTL expires and auto-releases seat lock; (3) *Network failure:* Idempotency keys prevent duplicate tickets on retry; (4) *Duplicate payment callback:* Idempotent webhook handler ignores repeat callbacks; (5) *Two users select same seat*

## 6. Master Role & Functionality Permission Matrix (PRD Reference)

28-MODULE RBAC MATRIX

The table below defines the complete 28-module access control specification across all four CineVerse roles (**Customer / User**, **Theatre Manager**, **Theatre Staff**, and **Super Admin**), distinguishing between standard usage, scoped configuration, operational monitoring, and administrative override rights.

| Platform Module (1–14) | User      | Manager     | Staff     | Admin       | Platform Module (15–28) | User | Manager        | Staff | Admin        |
|------------------------|-----------|-------------|-----------|-------------|-------------------------|------|----------------|-------|--------------|
| 1. Register / Login    | ✓         | ✓           | ✓         | ✓           | 15. Reviews             | ✓    | Respond        | ✗     | Moderate     |
| 2. Profile             | ✓         | ✓           | ✓         | ✓           | 16. Theatre Setup       | ✗    | ✓              | ✗     | Approve      |
| 3. Browse Movies       | ✓         | ✓           | ✗         | ✓           | 17. Screen Management   | ✗    | ✓              | ✗     | Override     |
| 4. Search              | ✓         | ✓           | ✗         | ✓           | 18. Seat Layout         | ✗    | ✓              | ✗     | Override     |
| 5. Movie Details       | ✓         | ✓           | ✗         | ✓           | 19. Show Creation       | ✗    | ✓              | ✗     | Override     |
| 6. Recommendations     | ✓         | ✗           | ✗         | Configure   | 20. Pricing             | View | Configure      | ✗     | Global Rules |
| 7. Theatre Discovery   | ✓         | ✗           | ✗         | ✓           | 21. Theatre Analytics   | ✗    | ✓              | ✗     | ✓            |
| 8. Book Tickets        | ✓         | ✗           | ✗         | ✗           | 22. Platform Analytics  | ✗    | ✗              | ✗     | ✓            |
| 9. Seat Selection      | ✓         | Configure   | ✗         | Configure   | 23. User Management     | ✗    | ✗              | ✗     | ✓            |
| 10. Smart Seat Rec.    | ✓         | ✗           | ✗         | ✗           | 24. Theatre Approval    | ✗    | ✗              | ✗     | ✓            |
| 11. Seat Locking       | Automatic | Monitor     | ✗         | Monitor     | 25. Coupons             | Use  | Theatre Offers | ✗     | Global       |
| 12. Payment            | ✓         | ✗           | ✗         | View        | 26. Staff Management    | ✗    | ✓              | ✗     | ✓            |
| 13. Cancellation       | ✓         | Limited     | ✗         | Override    | 27. Audit Logs          | ✗    | Limited        | ✗     | ✓            |
| 14. QR Ticket / Scan   | View QR   | View + Scan | Scan Only | View + Scan | 28. System Monitoring   | ✗    | ✗              | ✗     | ✓            |

### RBAC Implementation Guidelines & Scope Boundaries

SECURITY ARCHITECTURE

#### Token-Based Permission Enforcement (Zero Hardcoded Role Checks)

All API routes and UI guards must validate granular permission claims embedded in the user's session/JWT (`CREATE_MOVIE`, `CREATE_SHOW`, `MANAGE_SEATS`, `SCAN_TICKET`, `VIEW_ANALYTICS`, `MANAGE_USERS`) alongside tenant-scoping middleware (`theatre_id` ownership) rather than relying on coarse `if (role == "admin")` conditionals.

- Customer Scope (CUSTOMER):** Full self-service access to discovery, smart seat recommendation, automatic Redis seat locking, checkout, coupon usage, cancellations, reviews, and digital QR tickets; zero access to operational or administrative modules.
- Theatre Manager Scope (THEATRE\_MANAGER):** Strictly isolated to their assigned `theatre_id(s)`. Can configure screens, seat layouts, showtimes, tier pricing, theatre-specific offers, and staff accounts, respond to reviews, and monitor live seat locks and theatre analytics—cannot modify global movie metadata or alter completed payment ledgers.
- Theatre Staff Scope (THEATRE\_STAFF):** Minimalist operational footprint restricted to authentication, viewing today's assigned show roster, scanning QR tickets at the entry gate, and performing limited manual booking lookups by Booking ID or mobile number.
- Super Admin Scope (SUPER\_ADMIN):** Unfettered platform governance across master movie catalogues, theatre